

Computer Architectures

Exam of 18.09.2023 - part II

Question 1

Given any positive integer a , the operation of digitaddition generates a new positive integer $b > a$ by adding to a the sum of the digits of a . For example, if $a = 47$, the sum of the digits of a is $4 + 7 = 11$, and then $b = a + 11 = 58$.

The process can be repeated endlessly, forming a digitaddition series: 47, 58, 71, 79, 95,

Define an area of memory that will store the digitaddition series. Each element of the series is a word, and the area of memory must be large enough to store 50 elements.

Then, in the Reset_Handler, assign a value of your choice to the first element of the series.

You are requested to write two subroutines in ARM assembly language:

- 1) The `digitSum` subroutine receives in input a positive integer a and returns the value c , corresponding to the sum of the digits of a .

Important: a can have any number of digits (the upper limit is due to its representation as word).

Hint: You can obtain the digits of a number a (from the least significant to the most significant) by repeatedly dividing a by 10 and taking the remainder.

- 2) The `digitaddition` subroutine receives in input:
 - the address of an area of memory (where only the first value is initialized)
 - the number N of elements of the digitaddition series to be computed (including the first one)

The subroutine reads the first value saved in the area of memory. Then, it computes (and writes in that area of memory) all the subsequent elements of the digitaddition series. In the computation of each element, it calls the `digitSum` subroutine.

The `digitaddition` subroutine returns the sum of all the digits in the digitaddition series.

For example, if the initial element stored in the area of memory is 47 and $N = 5$, the subroutine stores in memory the next elements 58, 71, 79, 95. Finally, it returns $4 + 7 + 5 + 8 + 7 + 1 + 7 + 9 + 9 + 5 = 62$.

If there is an overflow in the computation of an element, the subroutine immediately ends and returns 0.

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Question 2

The sum of all the digits in a digitaddition series can be computed with a simple formula: you have to subtract the first number of the series from the last one and add the sum of the digits of the last number. For example, considering the series 47, 58, 71, 79, 95, the sum of all its digits is $95 - 47 + 9 + 5 = 62$. We want to check this formula in our program.

Add the following functionalities to the project created in the previous exercise:

- 1) The user can specify a binary value K through buttons Key1 and Key2. Key1 inserts a new digit equal to 0, while key2 inserts a new digit equal to 1. For example, if the user presses Key2, Key1, Key1, Key2, Key2, Key1, the final value is 100110
- 2) Declare an uninitialized array of int, with size 10.
- 3) When the user presses INT0:
 - set the first element of the array to K
 - call the `digitaddition` subroutine passing the address of the array and 10 (as the length of the series). The subroutine writes the other elements of the array and returns the sum of all the digits.
 - call the `digitSum` subroutine passing the last element of the array in order to obtain the sum of its digits.
 - If the value returned by the `digitaddition` subroutine is equal to the value returned by the `digitSum` subroutine plus the last element of the series minus the first element, then switch on led 4 and switch off led 5. If it is different, switch on led 5 and switch off led 4.